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Revitalizing Indigenous Knowledge Systems via Digital Media Technologies for Sustainability of Indigenous Languages

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Abstract: Amidst globalization and the Fourth Industrial Revolution, indigenous language preservation and promotion gain paramount significance. These languages encapsulate rich cultural, historical, and ecological value, embodying unique knowledge systems and identities. Nevertheless, challenges stemming from historical, social, resource, and institutional constraints impede preservation, with digital media technology thus emerging as a pivotal tool in this endeavour. This paper meticulously explores the role of digital media technology in indigenous language preservation during the Fourth Industrial Revolution. Through a comprehensive review of reputable literature, including scholarly articles, case studies, and expert insights, the paper underscores the imperative of language preservation, delving into the obstacles hindering preservation efforts and spotlighting the pivotal role of digital media technology in rejuvenating indigenous languages. The paper also examines various digital tools and applications that facilitate preservation, emphasizing their accessibility and engagement, while also underscoring the importance of collaborative efforts involving indigenous communities, linguists,

and technology experts. Ethical considerations, encompassing intellectual property rights and informed consent, are of paramount importance in indigenous language preservation, and thus this paper also consolidates insights derived from the findings and provides recommendations for effective preservation strategies, underlining the necessity of ethical guidelines and emphasizing the indispensable role of government support. The study concludes by addressing prospects and challenges in indigenous language preservation, providing a comprehensive view of safeguarding linguistic heritage in the digital age.

Keywords: indigenous language; preservation; digital media technology; Fourth Industrial Revolution

1 Introduction

Indigenous communities around the world possess a rich tapestry of cultural knowledge, traditions, and languages. The world is home to approximately 7000 languages, but a significant number of these are indigenous and face the threat of extinction; as indigenous communities adapt to modern life, their languages and knowledge systems are increasingly marginalized. These indigenous knowledge systems often passed down through generations are essential for the preservation of unique identities and the sustainable management of ecosystems, however, many indigenous languages are currently endangered, with their decline threatening the loss of vital knowledge. In this digital age, leveraging digital media technologies offers a promising avenue for the revitalization and preservation of indigenous knowledge systems and languages. This convergence leads to significant disruptions in traditional strategies and methods and, moreover, digital media technologies are driving transformative changes across various industries and institutions, reshaping how individuals and organizations interact and collaborate. These technologies are bringing about a revolution in our society by enabling the

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preservation and promotion of indigenous knowledge through tangible objects created with innovative materials. As digital media technologies represent a major era characterized by profound technological advancements reshaping global societies and industries, it is vital to acknowledge their influence (Oosthuizen, Ungerer, and Volschenk 2023). Amid this sweeping technological revolution, the preservation and revitalization of indigenous knowledge via digital media technologies emerge as a vital and pressing concern. Indigenous languages serve as crucial repositories of cultural heritage, embodying unique systems of communication, knowledge, and identity for diverse communities (Gwervevande and Mthombeni 2023).

However, the relentless forces of globalization and socio-cultural dynamics expose these practices to a genuine risk of extinction. In response, digital media technology assumes a central role, providing a dynamic and innovative toolkit to address the formidable challenges faced by indigenous languages. Digital media technology enables meticulous documentation, comprehensive archiving, and seamless digitization of indigenous languages, extending their accessibility to a broader and more diverse audience (Hinton et al. 2016). It also serves as a canvas for the creation of interactive and immersive language learning applications, online resources, and social media platforms designed to engage and empower indigenous communities. The true significance of digital media technology emerges as it bridges gaps between disparate communities, fostering cultural and linguistic diversity. By harnessing digital tools and platforms, speakers and advocates of indigenous knowledge can foster collaborations, share valuable resources, and revitalize their languages in unprecedented ways (Gwervevande and Mthombeni 2023).

As globalization and digital transformation reshape our world, understanding the profound cultural, historical, and ecological significance of indigenous languages becomes paramount. These languages are not merely repositories of linguistic diversity but also storehouses of unique knowledge systems and cultural identities; consequently, comprehending the importance of preserving indigenous knowledge is the linchpin for effective conservation efforts. In conjunction with acknowledging the value of indigenous knowledge preservation, it is equally crucial to identify and address the multifaceted challenges they face (Montero, Dénommé-Welch, and Henry 2022) which, stemming from historical factors, social dynamics, limited resources, and inadequate institutional support, present complex obstacles. Another challenge arises from the need to explore the evolving role of digital media technology in indigenous language preservation (Onyenankeya 2022). In an era where digital technologies and communication platforms have become ubiquitous, assessing

how they can be leveraged to protect and revitalize indigenous knowledge becomes a pressing issue.

This exploration encompasses the innovative use of digital tools and applications, enabling the documentation, archiving, and digitization of indigenous knowledge. It also entails the creation of interactive and engaging language learning applications and online resources, which are crucial elements in the preservation toolkit. In light of these complexities, effectively addressing these issues demands a thorough evaluation of the various digital tools and applications available for language preservation. The goal is to discern which tools are most effective in meeting the unique requirements of indigenous language preservation, with this task of particular significance as it guides the selection of tools that align best with the specific needs and challenges faced in this context. The sustainability of indigenous languages is intrinsically linked to the promotion of collaborative approaches that engage indigenous communities, linguists, and technology experts (Meighan 2023); determining the most effective collaborative strategies is thus essential for the success of preservation initiatives. This collaborative effort must also consider and address ethical considerations, such as intellectual property rights and informed consent, ensuring that resource utilization is responsible and respectful.

In subsequent discussions, the paper embarks on a comprehensive exploration of the multifaceted challenges and opportunities that underpin the preservation and promotion of indigenous languages through digital media technology. Overall, the paper also offers a unique perspective on the preservation and promotion of indigenous languages, leveraging digital media technology as a catalyst for change. Buttressing further, the paper sheds light on the critical role that digital media technologies play in preserving and revitalizing indigenous knowledge systems, particularly in the context of language sustainability. Furthermore, the paper also highlights the innovative strategies and collaborative efforts needed to preserve and promote indigenous languages. These insights, however, put the position of this paper in a unique direction, hence the need to bridge the gaps as observed in literature. Therefore, this paper will contribute to the growing body of knowledge in terms of policy, practice, and literature in the area of indigenous knowledge.

2 Relevance of Indigenous Knowledge Preservation

Indigenous knowledge is the living embodiment of cultural, social, and historical significance for Indigenous

communities worldwide. The cultural value of these practices lies in their ability to preserve and transmit ancestral knowledge, traditions, and oral histories, while indigenous knowledge also has a profound social value as it fosters community cohesion and identity. From a historical perspective, indigenous languages provide vital insights into the histories and experiences of indigenous communities. Recognizing the importance of indigenous languages, there are ongoing efforts to revitalize and preserve them, such as the United Nations Declaration on the Rights of Indigenous Peoples emphasizing the rights of Indigenous peoples to maintain, revitalize, and transmit their languages (Hohmann and Weller 2018). Indigenous knowledge, often referred to as traditional or local knowledge, is the collective wisdom and understanding that Indigenous communities have developed over generations. It encompasses a vast array of skills, practices, beliefs, and insights related to the environment, agriculture, health, spirituality, and social organization. Indigenous knowledge is deeply rooted in the cultures and traditions of these communities and has been passed down orally from one generation to the next.

The preservation of indigenous knowledge is not only essential for the cultural continuity of these communities but also holds significant relevance in our modern world (Masenya 2024). Preserving indigenous knowledge is a fundamental aspect of preserving the cultures and identities of indigenous communities. Indigenous knowledge is often intertwined with rituals, language, and spiritual beliefs that define a community's cultural identity, and as indigenous languages and traditions are threatened by globalization and modernization, the preservation of indigenous knowledge becomes a vital tool in safeguarding these cultures. It serves as a means of cultural resistance against the homogenizing forces that can erode unique identities.

Indigenous communities have, for centuries, lived in harmony with their natural environments, with their knowledge of local ecosystems, sustainable agriculture, and resource management invaluable in addressing contemporary environmental challenges. Indigenous knowledge provides insight into how to mitigate climate change, protect biodiversity, and preserve ecosystems. For instance, the traditional ecological knowledge of indigenous peoples can inform sustainable agricultural practices and the conservation of endangered species; preserving this knowledge is crucial for promoting responsible environmental stewardship. Indigenous knowledge systems often include a rich repository of traditional medicines and healing practices (Otang-Mbeng et al. 2023), with these systems based on extensive knowledge of local flora and fauna and their medicinal properties. The importance of this knowledge is increasingly recognized as it offers alternative or

complementary approaches to modern medicine; moreover, the development of new pharmaceuticals and treatments can benefit from indigenous knowledge. For instance, drugs like quinine, derived from the cinchona tree, were originally used by indigenous people in South America to treat malaria (Bird et al. 2019).

3 Role of Digital Media Technology in Preserving Indigenous Knowledge

Digital media technology plays a pivotal role in supporting the preservation, documentation, and revitalization of indigenous languages, offering a diverse array of tools and platforms that can significantly enhance language learning, facilitate documentation and archiving efforts, and broaden accessibility to indigenous languages. Digital media technology furnishes robust tools for the documentation and archiving of indigenous languages, such as utilizing audio and video recording devices alongside digital storage and preservation systems which enables the capture and preservation of oral traditions, songs, stories, and other linguistic resources in high-quality formats (Hinton 2019). The creation of digital archives ensures the long-term preservation and accessibility of indigenous language materials for future generations, mitigating the risk of their loss due to physical deterioration or cultural disruption.

Furthermore, digital media technology facilitates the digitization of existing knowledge resources, including printed materials, manuscripts, and linguistic analysis materials. This digitization process preserves these resources in a more accessible and shareable format, thereby streamlining their dissemination among researchers, educators, and community members (Bird et al. 2019). Additionally, digital media technology offers innovative approaches to learning and teaching indigenous knowledge, with mobile applications, online platforms, and interactive software providing opportunities for self-paced language learning, vocabulary acquisition, and grammar instruction (Austin 2019). These tools often incorporate multimedia elements, such as audio recordings, video tutorials, and interactive exercises, which significantly enhance engagement and interactivity in the learning process. Importantly, digital knowledge learning resources can reach a wider audience, including individuals living in remote areas or outside of indigenous communities.

Moreover, digital platforms, particularly within the realm of social media, offer opportunities for language promotion and community engagement. Indigenous

language speakers can establish online communities, social media groups, and dedicated websites for sharing language resources, engaging in discussions about language-related topics and providing support for language learners (Molnar and Chartrand 2021). These platforms effectively foster connections among speakers, learners, and language enthusiasts, creating a sense of belonging and encouraging collective efforts toward indigenous language revitalization. Digital media technology also empowers collaborative knowledge documentation initiatives, with online platforms serving as facilitators for collective endeavours in knowledge documentation, enabling the amalgamation of linguistic expertise, cultural wisdom, and community contributions. Various apps and software tools dedicated to revitalization have emerged to bolster the preservation of indigenous knowledge; these resources frequently encompass elements like pronunciation guides, language-based games, flashcards, and interactive exercises, which actively engage learners in language acquisition (Harmon 2019). Language revitalization apps offer user-friendly interfaces that ensure accessibility for individuals interested in indigenous languages, regardless of their location or language proficiency.

The versatility of digital media technology permits the creation of multimedia materials aimed at reinvigorating indigenous languages, which includes the production of audiovisual content like podcasts, videos, and interactive storytelling apps (Dutta et al. 2019). Multimedia formats enrich the language learning experience by incorporating visual and auditory stimuli, making knowledge acquisition a more engaging and memorable process for learners, which is also achieved by digital media technology introducing opportunities for gamification within the realm of language learning. Language acquisition apps and platforms can seamlessly integrate game elements such as challenges, rewards, and progress tracking, thereby motivating learners (Beermann and Spreer 2018); this gamification strategy nurtures a lively and interactive environment, encouraging language engagement and facilitating knowledge retention. Through digital platforms, indigenous communities can document and share their traditional ecological knowledge, cultural practices, and medicinal plant information (Adam, Othman, and Halim 2022). This not only preserves and safeguards indigenous knowledge for the benefit of future generations but also contributes to broader cultural heritage preservation initiatives.

The accessibility and global reach afforded by digital platforms have revolutionized the promotion of indigenous knowledge, surmounted geographical barriers, and made it accessible to a worldwide audience. Online platforms enable the dissemination of traditional materials, recordings, and

learning resources, significantly enhancing their visibility and impact. Indigenous communities now can harness digital media technology to showcase their languages, culture, and traditions, raising awareness and nurturing a greater appreciation for linguistic diversity (Dutta et al. 2019). However, it is imperative to acknowledge the challenges and limitations that accompany the use of digital media technology in indigenous knowledge preservation. Issues such as limited access to technology and internet connectivity can impede the reach and effectiveness of digital resources in certain communities. Consequently, it is crucial to address these challenges and strive to bridge the digital divide, ensuring that efforts are made to render digital resources more accessible to all (Beermann and Spreer 2018).

Indigenous communities should have the agency to determine how their traditional practices are represented in the digital realm, ensuring cultural protocols and ethical considerations are upheld (McIlraith et al. 2020), which involves actively involving community members in the development and implementation of digital media initiatives, ensuring that they align with the specific needs and goals of the community. Despite these challenges, the integration of digital media technology in indigenous knowledge preservation presents a transformative opportunity to revitalize languages, engage learners, and foster connections within the community, as the use of digital tools enables interactive and immersive language-learning experiences, empowering individuals to take ownership of their language-learning journey (Lotherington and Reinhardt 2019). Digital platforms also facilitate community networking and collaboration, connecting indigenous knowledge speakers, educators, and researchers from different parts of the world, allowing for the exchange of knowledge, resources, and best practices (Scholze-Irrlitz and Struß 2017).

4 Digital Tools and Applications for Indigenous Knowledge Preservation

In recent times, a plethora of digital tools have emerged as influential resources for language preservation, offering innovative avenues for documentation, learning, and revitalization efforts, and these technological advancements have significantly enhanced knowledge preservation, making it more accessible and engaging for both linguists and language communities. The advent of digital tools, including knowledge documentation software, archiving systems, text analysis software, language learning applications, and language translation software, along with various online

applications and social media platforms, has reshaped the landscape of language preservation. Mobile apps, in particular, have gained widespread popularity as essential tools for language learning. Many of these apps now offer courses and resources specifically tailored to endangered and indigenous languages.

Social media platforms have played a pivotal role in uniting communities around endangered languages. Dedicated pages or channels on platforms such as Facebook, Instagram, and YouTube serve as virtual hubs where speakers of endangered traditions can share content and stories and engage with others who are passionate about learning and preserving their language. These online communities foster traditional practices, cultural exchange, and networking opportunities, contributing to the revitalization of these linguistic treasures (Living Tongues Institute n.d.). Moreover, the introduction of virtual reality (VR) and augmented reality (AR) technologies has ushered in immersive traditional learning experiences, providing learners with an interactive and dynamic platform to engage with indigenous knowledge.

Through VR and AR, learners simulate real-life traditional scenarios, such as conversations with native speakers or virtual field trips to language-speaking regions, with these technologies offering an interactive and engaging environment for learners to practice and engage with endangered languages (Sweller et al. 2019). Online forums, discussion boards, and chat platforms dedicated to specific languages have also become valuable resources; these platforms create virtual communities where language learners and speakers can connect, ask questions, and share resources, with such online communities fostering traditional practice, peer support, and networking among individuals interested in preserving endangered traditions (Living Tongues Institute n.d.). Additionally, specific apps like SayMore, ELAN, and FLEEx have been developed to assist linguists and researchers in the collection, transcription, and analysis of language data in the field (Living Tongues Institute n.d.); these apps streamline the documentation process and ensure the preservation of linguistic information for future reference and analysis. Websites like the Living Tongues Institute for Endangered Languages serve as centralized hubs for language preservation efforts, providing resources, multimedia materials, and information on endangered languages, supporting documentation, revitalization initiatives, and educational efforts (Living Tongues Institute n.d.). By leveraging these digital tools and applications, successful initiatives have been able to reach broader audiences, engage communities, facilitate language learning, and ensure the long-term preservation of endangered languages. These initiatives exemplify the potential of digital media technology in supporting language preservation efforts.

In the digital age, the preservation of indigenous knowledge is increasingly recognized as a vital endeavor to safeguard cultural heritage and promote sustainable development. Indigenous communities especially in Africa possess a wealth of traditional knowledge, passed down through generations, encompassing diverse domains such as medicine, agriculture, ecology, and storytelling. In so doing, Balogun and Kalusopa (2021) argue that innovative digital tools and applications are being leveraged to document, archive, revitalize, and transmit indigenous knowledge in Africa. Going further, the establishment of the Indigenous Digital Archive (IDA) in South Africa was another remarkable initiative in the projection of using digital technologies for traditional knowledge perseverance. Mogajane (2022) adds that the Indigenous Digital Archive project in South Africa aims to preserve and promote indigenous languages and knowledge through digital platforms, which is to say that the indigenous digital archive collaborates with local communities to digitize archival materials, including oral histories, songs, and cultural artifacts. While making available these resources accessible in online platforms, the IDA facilitates local knowledge transmission and revitalization within indigenous communities as well as raises awareness among a broader audience about the richness of their cultures.

Exploring further, the Sankofa Digital Archive initiative in Ghana focuses on preserving the cultural heritage of the Akan people through digital technologies (Adu and Ngulube 2016). The authors submit that the initiative involves digitizing traditional stories, proverbs, and rituals, as well as documenting indigenous practices related to agriculture, craftsmanship, and spiritual beliefs. It was noted that the Sankofa Digital Archive provides training in digital literacy and encourages active participation from indigenous communities in the preservation process from partnerships with local universities and community organizations.

In many African countries, mobile applications such as “Ntongato” in Tanzania and “Learn Yoruba” in Nigeria have been developed to facilitate language learning and revitalization (Mogajane 2022). These apps offer interactive lessons, vocabulary quizzes, and audio recordings of native speakers, making language learning accessible and engaging for both indigenous community members and learners from diverse backgrounds. On the other hand, Fuseini (2021) underscores that indigenous communities in Africa possess intricate knowledge of local ecosystems, including biodiversity hotspots, migration patterns, and sustainable land management practices, with GIS technology used for mapping and visualization of indigenous environmental knowledge which enable communities to document changes in their natural surroundings, identify areas of ecological importance, and

advocate for conservation measures. For example, the “Community Mapping for Environmental Management” project in Kenya empowers Maasai communities to use GIS tools to map grazing areas, water sources, and wildlife corridors, contributing to the sustainable management of their ancestral lands.

Following the fact that storytelling plays a central role in indigenous cultures and serves as a means of preserving history, transmitting knowledge, and fostering cultural identity, digital storytelling platforms such as “StoryMap” and “VoiceThread” offer indigenous communities a multi-media toolkit for sharing their oral traditions, folklore, and personal narratives with a global audience (Adu and Ngulube 2016). Aiseng (2023) states that digital platforms enable indigenous storytellers to create immersive digital narratives that celebrate their cultural heritage and promote cross-cultural understanding; digital tools and applications thus have immense potential to support indigenous knowledge preservation efforts in the African context and beyond. Given the power of technology, indigenous communities can document, archive, and transmit their traditional knowledge for future generations which foster cultural resilience and promote intergenerational learning.

5 Collaborative Measures Towards the Sustainability of Indigenous Knowledge

The collaboration between indigenous communities and technology experts stands as a paramount element in the endeavors to preserve knowledge. Through this collaborative approach, a fusion of diverse skills and knowledge is achieved, resulting in the success of preservation initiatives. Indigenous communities, with their rich traditional and cultural wisdom, offer invaluable insights and perspectives that are fundamental to the accurate preservation of knowledge; their active involvement is essential to ensure the fidelity of preservation efforts. In complementing the contributions of indigenous communities, technology experts play a pivotal role by harnessing their expertise to craft digital tools and applications that facilitate knowledge preservation. Their contributions encompass a wide spectrum of activities, ranging from the creation of user-friendly interfaces to the development of mobile apps and online platforms dedicated to language learning and documentation (Liu et al. 2019). Supporting this, Aiseng (2023) states that collaboration with technology partners enables the digitization and dissemination of indigenous knowledge through digital platforms, mobile applications, and online repositories. With the power of

digital technologies, collaborative initiatives enhance the accessibility, visibility, and preservation of indigenous knowledge for future generations; these technological innovations serve to enhance the accessibility and engagement associated with knowledge preservation initiatives.

Partnerships between academia, indigenous organizations, and tech companies have yielded successful knowledge preservation projects. Academia provides research expertise, funding opportunities, and access to resources like libraries and archives. Indigenous organizations bring cultural knowledge, community connections, and a deep understanding of the language and its context. Tech companies contribute to technological advancements, resources, and wider reach in terms of user adoption and distribution of knowledge preservation tools (Hinton 2014). Indigenous communities should actively participate in defining project goals and shaping the direction of initiatives (Reyhner and Lockard 2009), with such community involvement fostering ownership and empowerment while strengthening cultural identity and revitalization efforts (McCarty et al. 2017); the foregoing further emphasizes the place of collaboration towards sustaining indigenous knowledge. Balogun and Kalusopa (2021), in the underlisted items, explain the essence of collaboration in the preservation of indigenous knowledge. Some of these include:

- Collaborative efforts are essential for preserving and revitalizing traditional knowledge systems. This is to further state that collaborative initiatives contribute to the preservation of cultural heritage and promote intergenerational transmission of knowledge through documenting of oral histories, recording of indigenous languages, and safeguarding of cultural artifacts.
- Collaborative projects that integrate indigenous knowledge with modern science and technology may lead to innovative solutions for addressing environmental challenges and promoting sustainable development in Africa.
- Collaborative measures empower indigenous communities to assert their rights, participate in decision-making processes, and safeguard their cultural heritage.
- Collaborative efforts foster dialogue and mutual understanding between indigenous and non-indigenous stakeholders, promoting respect for cultural diversity and indigenous rights.
- Collaborative initiatives contribute to building inclusive and harmonious societies in Africa through creating platforms for knowledge exchange and cross-cultural learning.

Aside from the collaborative efforts of other organizations and institutions towards the sustainability and preservation

of indigenous knowledge, Sithole (2007) states that collaboration between libraries and local communities is essential for ensuring that preservation efforts are culturally sensitive, respectful, and responsive to the needs and aspirations of indigenous peoples. Sarkhel (2017) also adds that libraries collaborate by consulting with local indigenous communities to understand their priorities, concerns, and aspirations regarding the preservation of their knowledge and cultural heritage. As highlighted by Reijerkerk and Nyitray (2023), libraries could collaborate with local communities to preserve and manage indigenous culture and traditions in the following ways:

- Libraries can work with local communities to document and archive indigenous knowledge, including traditional stories, oral histories, cultural practices, and artifacts. Collaborative efforts may involve conducting interviews with community elders and knowledge holders, digitizing archival materials, and creating digital repositories accessible to the community and researchers.
- Libraries can organize cultural programming and events in collaboration with local indigenous communities to showcase and celebrate their cultural heritage. This may include storytelling sessions, language classes, traditional music and dance performances, art exhibitions, and cultural festivals.
- Libraries can support language revitalization initiatives led by local indigenous communities by offering resources and programs aimed at preserving and promoting indigenous languages. This may include bilingual books, language learning materials, language immersion programs, and literacy initiatives tailored to the needs of the community.
- Libraries can engage in community-based research projects in collaboration with local indigenous communities, exploring topics related to traditional ecological knowledge, cultural practices, and local histories. Collaborative research efforts involve indigenous researchers, scholars, and community members in the co-creation of knowledge, ensuring that research outcomes are culturally sensitive and relevant to the needs of the community.
- Libraries can establish cultural heritage centers or indigenous knowledge hubs within their premises, providing dedicated spaces for preserving and showcasing indigenous knowledge. These centers serve as community gathering places where indigenous elders, knowledge holders, and youth can gather to share their wisdom, conduct workshops, and engage in intergenerational learning.

6 Future Expectations and Challenges

Emerging technologies hold remarkable potential for advancing knowledge-preservation endeavors. These cutting-edge technologies introduce novel approaches to enhance knowledge documentation and foster cultural immersion, thereby opening up fresh opportunities for indigenous communities. Artificial intelligence (AI) plays a pivotal role in knowledge preservation by facilitating the development of knowledge-learning applications and tools. It harnesses the power of AI to create innovative solutions that empower indigenous communities in their preservation efforts.

Virtual reality (VR) technology represents a particularly promising avenue for creating immersive traditional experiences. VR can transport knowledge to virtual environments where individuals can engage with native speakers, hone their language skills, and immerse themselves in cultural contexts. Furthermore, VR technology can be utilized to digitally preserve cultural heritage sites, traditional practices, and storytelling, thereby offering future generations a means to connect with their cultural heritage and knowledge (García-Peñalvo et al. 2020). However, it is essential to recognize the challenges that arise concerning the digital divide and access to technology within indigenous communities. The digital divide refers to the disparities in technology access and internet connectivity, which can pose obstacles to knowledge preservation initiatives. Many indigenous communities, especially those residing in remote areas, may encounter limitations related to infrastructure, lack of internet accessibility, and inadequate technology resources (Dutta et al. 2019).

To address these challenges, it is crucial to ensure equitable access to technology and bridge the digital divide. Governments, organizations, and tech companies can collaborate to provide necessary infrastructure, internet connectivity, and digital devices to indigenous communities. Initiatives should be implemented to promote digital literacy and skills training among community members, empowering them to effectively utilize emerging technologies for knowledge preservation (United Nations 2021). Furthermore, community involvement and participatory approaches are vital for addressing the challenges related to technology access. Indigenous communities should have a say in the design and implementation of preservation projects, ensuring that the technologies used align with their cultural values, needs, and preferences; such collaborative partnerships between indigenous communities, researchers, and technology experts can

foster culturally sensitive and community-driven approaches (Turin 2021). The long-term sustainability of knowledge preservation efforts relies on continuous support from various stakeholders, with traditional preservation not a one-time task but a continuous process that requires ongoing commitment and resources. Studies have revealed the factors contributing to the long-term sustainability of the preservation of indigenous knowledge:

- (1) Sustainable financial resources play a pivotal role in ensuring the continuation of knowledge preservation initiatives. Governments, philanthropic organizations, and other stakeholders must allocate dedicated funding for language preservation projects; these financial resources can provide essential support for a wide array of activities, including traditional documentation, curriculum development, technology infrastructure, and community engagement. Sustained funding serves as a guarantee for the longevity of these initiatives, enabling them to tackle challenges such as training indigenous experts, developing educational materials, and maintaining digital platforms (Hinton et al. 2016).
- (2) The active engagement of indigenous communities stands as a linchpin for the enduring sustainability of indigenous preservation efforts. Communities must be empowered to assume ownership of knowledge preservation initiatives and actively participate in the decision-making processes. This engagement instils a sense of ownership, motivating community members to contribute actively to language revitalization and ensuring that preservation efforts align with their cultural values and needs. Sustainable traditional preservation relies on ongoing collaboration, communication, and capacity building within indigenous communities (Grenoble and Whaley 2006).
- (3) Emphasis on traditional knowledge revitalization initiatives should be placed on education and transmission, particularly targeting younger generations. The intergenerational transmission of traditional knowledge is a cornerstone of cultural sustainability, with initiatives that seamlessly integrate traditional preservation into formal and informal educational settings, such as schools, community centres, and digital learning platforms, having the potential to ensure the continued passage of indigenous traditions to forthcoming generations. Comprehensive programs encompassing immersion schools, cultural nests, and mentorship initiatives can significantly facilitate the acquisition and perpetuation of traditional knowledge (McCarty et al. 2016).
- (4) The integration of technology and digital tools plays a pivotal role in securing the long-term sustainability of

traditional preservation endeavours. Digital platforms, mobile applications, and online resources offer accessible and interactive means for learning, practising, and documenting indigenous knowledge. These technological tools enable the broader dissemination of traditional materials, foster collaboration between linguists and indigenous communities, and contribute to the preservation and revitalization of traditions in the digital era. Continuous technological support and adaptation to emerging technologies are fundamental for ensuring the enduring impact of traditional preservation efforts (Bird and Simons 2003).

- (5) Governments and policymakers hold a central role in creating an enabling environment for traditional preservation. Legislative measures may encompass the recognition of oral traditions, the integration of indigenous languages into education and public services, and the establishment of local knowledge preservation institutions. Robust policy frameworks serve as a solid foundation for the long-term sustainability of languages and traditions (UNESCO 2003).

7 Conclusion and Implication of the Study

The culmination of this study underscores the profound significance of preserving and promoting indigenous knowledge, which extends beyond the cultural heritage of indigenous communities to impact their overall well-being. These traditions serve as invaluable repositories of knowledge systems, cultural expressions, and unique worldviews that contribute to the rich tapestry of human diversity. Moreover, the transformative role of digital media technology in revitalizing and advocating for indigenous knowledge cannot be overstated; the importance of safeguarding indigenous knowledge goes far beyond mere historical preservation. These traditions serve as vessels carrying the legacies, histories, and collective wisdom of communities that have thrived for generations. They represent distinct worldviews intricately connected with the environment, spirituality, and identity of their speakers. This study reaffirms that indigenous knowledge is not confined to being relics of the past; instead, they remain living repositories of human understanding and cultural profundity.

In this modern age, digital media technology emerges as the proverbial torchbearer for indigenous knowledge preservation. It offers innovative and powerful tools and platforms that are instrumental in the revitalization and propagation of these traditions. Through the lens of digital

media, we can witness the creation of digital archives, mobile applications, and online resources that facilitate the documentation, learning, and dissemination of indigenous knowledge. These technologies transcend barriers and provide accessible and interactive avenues for individuals, whether they belong to indigenous communities or not. A significant novelty arising from our study is the realization that digital media technology is not just a means of preservation but a powerful catalyst for knowledge revitalization. Furthermore, the study has illuminated the capacity of digital media technology to comprehensively document traditions, ensuring the survival of endangered cultural practices. As a result, these traditions are now well-prepared to thrive, adapt, and evolve in the digital age, securing their enduring place in the ever-evolving cultural tapestry of the future. Moreover, libraries have played a crucial role in facilitating collaboration by consulting with local communities, organizing cultural events, supporting language revitalization initiatives, engaging in community-based research, and establishing cultural heritage centers. Moving forward, it is essential to continue fostering collaboration between all stakeholders involved in indigenous knowledge preservation.

8 Recommendations

Arising from the discussion in this study, it is recommended that:

- (1) It is imperative to prioritize community-led initiatives that actively involve indigenous communities in the preservation process. This entails consulting with community members, elders, and knowledge holders to identify priorities, concerns, and strategies for preserving indigenous knowledge.
- (2) There is a need to strengthen partnerships between libraries, academia, technology companies, and indigenous organizations to leverage collective expertise, resources, and networks.
- (3) There should be adequate investments in digital literacy training and infrastructure development that can empower communities to actively participate in digital preservation initiatives and ensure the long-term viability of indigenous knowledge in the digital age.

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